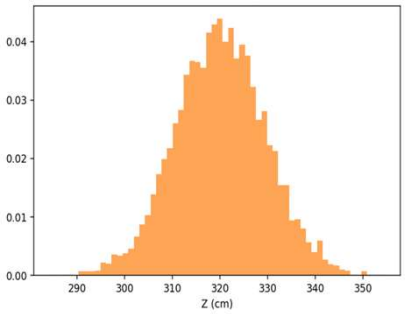
Agenda

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Last time:
  - z-score and proportions
  - Sums of independent but not identical normal distribution
  - Central Limit Theorem
    - "Common man" version
    - Technical version

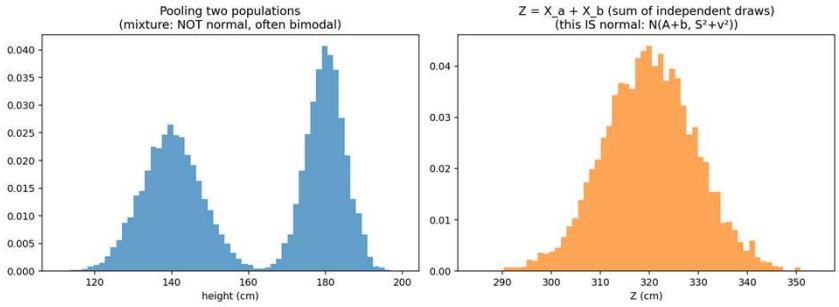


2

Sums of independent normal distribution

- $Y = X_1 + X_2 + \cdots X_n$ where each $X_i \sim N(\mu_i, \sigma_i^2)$ and independent
  - Answer: $Y \sim N(\sum_i \mu_i, \sum_i \sigma_i^2)$
- Per theorem, what happens if measure heights X_1 of national basketball players and, say, kho-kho players X_2?

The result is not about concatenation, or mixture



3

Why normal?

- For independent identical (iid) random variables, and $Y = X_1 + X_2 + \cdots X_n$
- Central limit theorem
  - For large (but finite) n, the distribution of Y is approximately normal with mean and variance as earlier seen as if the distributions were normal
  - Finite n, sample mean $\overline{x} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
    - Take home message emphasizes the division by n
    - Although Y above is actually Y_n

Take home message

4

Normal and Sums of Bernoulli

- Let X= $X_1 + X_2 + \cdots X_n$
  - Each $X_i \overset{\text{iid}}{\sim} \text{Bern}(p)$
- Clearly $X \sim \text{Bin}(n, p)$
  - We know exact distribution
- Central limit theorem approximation:
  - For large n, $X \sim \mathcal{N}(np, np(1-p))$
  - $\frac{\text{X}-np}{\sqrt{np(1-p)}}$ is standard normal

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

5

Central limit theorem

More precise statement

- $X_1, X_2, \ldots, X_n$ independent and identically distributed random variables each with mean μ and variance $\sigma^2 < \infty$
  - $Y_n \triangleq X_1 + X_2 + \cdots X_n$
- As $n \to \infty$, the distribution Y_n converges to that of $\mathcal{N}(0, \sigma^2)$
  - Y_n converges in distribution to normal

The statement is more precise but a tad awkward

Ignore

Typo:

See next slide

Convergence in distribution (see here for more)

$$Y_n = \sqrt{n}\left(\frac{\sum_{i=1}^{n} X_i}{n} - \mu\right)$$

78

6

CLT

Why the complicated statement?

$$Y_n = \sqrt{n}\left(\frac{\sum_{i=1}^{n} X_i}{n} - \mu\right) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

Untangling

$$Y_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \dot{\sim} N(0, \sigma^2)$$

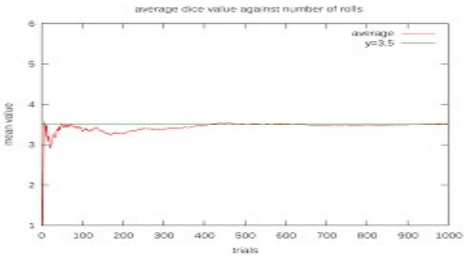
$$\frac{1}{n}\sum_{i=1}^{n} X_i - \mu \;\dot{\sim}\; N(0, \sigma^2/n) \qquad X \sim N(\mu, \sigma^2) \text{ and } Y = aX, \implies Y \sim N(a\mu, a^2\sigma^2)$$

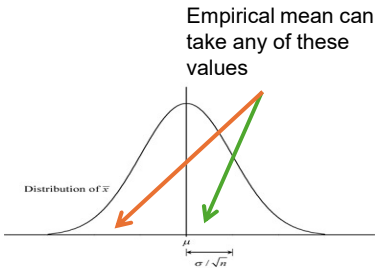
$$\frac{1}{n}\sum_{i=1}^{n} X_i \;\dot{\sim}\; N(\mu, \sigma^2/n)$$

7

CLT and LLN

- The law of large numbers says that the empirical mean calculated from a large number of samples is equal to (or very close) to the true mean μ (of the distribution from which the samples were drawn)
- The central limit theorem says that the empirical mean calculated from a large number of samples is a random variable following a Gaussian distribution with mean equal to the true mean μ



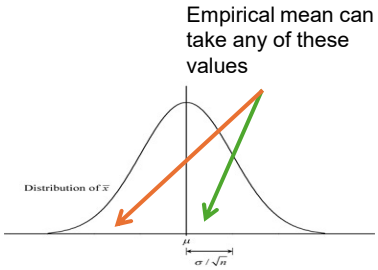


83

8

CLT and LLN

- Limiting argument as $n \to \infty$
  - The variance drops to zero



$$Y_n = \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right) \dot{\sim} N(0, \sigma^2)$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i - \mu \;\dot{\sim}\; N(0, \sigma^2/n) \qquad X \sim N(\mu, \sigma^2) \text{ and } Y = aX, \implies Y \sim N(a\mu, a^2\sigma^2)$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i \;\dot{\sim}\; N(\mu, \sigma^2/n)$$

83

9

The weak law of large numbers

- Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables, each with mean μ. For any $\varepsilon > 0$:

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \text{ as } n \to \infty$$

- The **sample mean** converges in probability to the true mean

Must know

10

10

CLT

- As stated, the random variables X_i are independent and identically distributed.
- Alternate: require independence but allow the random variables to belong to different distributions
  - Technical conditions (e.g. variance of no one random variable, or groups of random variables is allowed to dominate the sum of the variances)
  - "take home message" … it works
- Misconceptions
  - "Lots" of random sampling "reveals" normal distribution
  - Does not apply to random sampling of any variable
    - Applies to sampling distribution of mean
  - Empirical rule of thumb (n=30) can lead to seriously flawed inferences

79

11